

Shweta Singh

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[LeetCode](#) | [Codeforces](#) | [Portfolio](#)

Education

J.K. Institute of Applied Physics and Technology, University of Allahabad	2024 – 2028
<i>Bachelor of Technology (B.Tech) in Computer Science and Engineering (AI Major) Current SGPA: 7.80/10</i>	
Class XII (Senior Secondary Schooling)	89 percent
Class X (Secondary Schooling)	93 percent

Experience and Contributions

GirlScript Summer of Code (GSSoC 2026)	May 2026 – Present
<i>Open Source Contributor (Full-Stack Engineering and Version Control)</i>	

- Contributed to production-ready open-source repositories by identifying structural layout anomalies, debugging modular code blocks, and optimizing codebase architectural compliance.
- Collaborated with maintainers and remote developer cohorts using advanced Git version control workflows, managing branches, resolving merge conflicts, and initiating pull requests.
- Refactored client-side interface components to optimize asynchronous asset rendering speeds and enhanced comprehensive developer documentation across multi-layered software modules.

Projects

ParkVision-Saathi – AI-Powered Parking Enforcement Intelligence	Jun 2026
<i>Python, FastAPI, ML, Game Theory</i> 🔗 GitHub 🔗 Live Demo	

- Engineered an AI-driven parking enforcement intelligence system for Bengaluru Traffic Police, quantifying illegal parking violations using a proprietary 6-factor Congestion Impact Score (0–100) per zone.
- Built predictive analytics modules to forecast parking violation hotspots, enabling proactive patrol deployment and reducing parking-induced congestion by up to 25 percent.
- Implemented Stackelberg game theory optimization combined with waterbed spillover simulation to determine optimal patrol team allocation across limited resources.
- Delivered a scalable, open-source prototype (MIT License) in 3 days during a hackathon, demonstrating rapid prototyping and community-focused innovation aligned with smart city goals.

SENTINEL – Autonomous Spacecraft FDIR Agent	Jun 2026
<i>Python, RAG, LLM, Anomaly Detection</i> 🔗 GitHub 🔗 Live Demo	

- Developed a Gemini-first, model-agnostic spacecraft safe-mode diagnosis and recovery system automating FDIR for ground operators, reducing manual telemetry analysis time by 80 percent.
- Implemented statistical Z-score anomaly detection on raw crash dumps and pre-fault telemetry to identify root causes of spacecraft failures in real time.
- Built a Hybrid RAG pipeline over ECSS standards to retrieve standard FDIR procedures, enabling grounded, auditable reasoning for safety-critical recovery decisions.
- Designed an interactive rich UI featuring a Causal DAG and Risk-Assessed Recovery Plan, streaming auditable reasoning traces for mission-critical transparency.

NagarSetu – AI-Driven Civic Architecture Platform	Dec 2025
<i>React.js, Node.js, Docker, GCP, MongoDB</i> 🔗 GitHub 🔗 Live Demo	

- Engineered a full-stack, REST-compliant web application for real-time tracking, sorting, and automated cloud reporting of regional civic infrastructure anomalies.
- Containerized the entire application stack using Docker to build isolated development environments, streamlining deployment pipelines and ensuring cross-platform stability.
- Implemented asynchronous geo-tagging pipelines and dynamic location telemetry workflows using Google Maps API, tracking coordinates with high spatial precision.
- Integrated Google Vision API on Google Cloud Platform to run image classification models, enabling automated validation and visual categorization of submitted user issues.

- Architected a backend priority-sorting algorithm that slashed manual administrative triage overhead and accelerated municipal response metrics by over 40 percent.
- Commanded a 4-member agile development squad, spearheading sprint planning, architectural design, and final competitive technical pitches.

Technical Skills

Programming Languages: Python, C, C++, JavaScript (ES6+), SQL, HTML5, CSS3, LaTeX

AI and Machine Learning: TensorFlow, Scikit-learn, OpenCV, Neural Networks, Predictive Modeling, Data Preprocessing, RAG, LLM Integration, Anomaly Detection, Game Theory

Frameworks and Web Technologies: React.js, Node.js, Express.js, FastAPI, TypeScript, REST APIs, Bootstrap

Cloud, DevOps and Engineering: Docker, Google Cloud Platform (GCP), System Design (Microservices), Git, Linux

Core Engineering Coursework: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems

Achievements

- **Top 10 National Finalist:** Selected among thousands at Square Hacks 2.0, IIT Delhi during the student exchange initiative.
- **National Runner-up:** Earned podium ranking at TechJam 2.0 held at Microsoft Noida Headquarters.
- **Top 13 Winner:** Evaluated in elite standing at CODEBLOCK Hackathon (University of Allahabad) for outstanding AI and Blockchain architecture.
- **City Topper:** Attained top rank in Technothon, administered by IIT Guwahati.
- **Campus Leadership Role:** Appointed as corporate relations Campus Ambassador for Renaissance 2025 via E-Cell MNNIT.
- **Varsity Athlete:** Represented university in State-level and North Zone Inter-University Tournaments for Basketball and Kabaddi.